MESH: Multi-receiver Evaluation of Spatial Hearing across Egocentric and Allocentric Views

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Abstract

Spatial hearing in multi-receiver environments requires identifying who speaks, where they are, and how they move. Existing benchmarks evaluate spatial prediction, transcription, tracking, and distributed speech processing, but offer limited evaluation of whether speech signal remains associated with the correct physical source across receivers and motion. We introduce MESH, which evaluates this association through egocentric and allocentric views of shared acoustic scenes. The egocentric view uses local recordings and peer metadata, while the allocentric view combines synchronized recordings from multiple receivers in a shared 3D world frame. MESH contains nearly 40,000 questions across 13 static and dynamic localization and spatial transcription tasks. We evaluate eleven audio-model configurations and a language-model agent using specialist tools. The best static distance-estimation and dynamic direction-tracking accuracies are only 21.90% and 41.54%, respectively, while the lowest word error rates across four transcription tasks range from 59.15% to 73.10%. Ablation studies and additional experiments reveal that existing models are not yet able to effectively leverage peer information and struggle to handle multi-speaker and dynamic scenarios. These findings highlight the gap between access to distributed acoustic information and its effective use for spatial speech understanding. We release the dataset and evaluation code to support further research on collaborative and distributed spatial hearing.[1]

1 Introduction

Real acoustic environments contain multiple sound sources, some stationary and others moving, that jointly form a complex and changing acoustic scene. Their emissions overlap with environmental noise and reverberation, making it difficult to determine what each speaker says, where they are, and how they move. We consider a distributed listening setting in which multiple receivers equipped with microphone arrays observe this scene from different locations. In embodied applications, these receivers may also move and emit sound, making them both observers and participants in the acoustic environment. Each receiver captures spatial evidence from its own position, while shared receiver poses, known playback signals, and recordings from other locations provide additional information about the same scene. This setting motivates two coordinate-based views: understanding surrounding sources in an individual receiver's local frame, and understanding source locations and speech in a shared 3D world frame. Peer collaboration and distributed recordings provide information for these views. An exhibition hall provides a concrete example: guide robots operate among visitors who stand near exhibits or walk between displays, while visitors and robots may speak simultaneously. A listening system must associate each utterance with the correct visitor, locate that visitor, and follow their motion as the relative geometry changes.

Existing benchmarks evaluate important components of this problem, including numerical spatial prediction and spatially selective transcription (Liu et al., 2026a; Zhu et al., 2026; Kumar et al., 2026), while distributed-recording resources support speech processing across devices (Schmalenstroeer et al., 2023). However, performance on an individual task does not establish that a system can correctly associate speech, location, and motion in a shared scene: a correct direction estimate may leave distance unresolved, and a fluent transcript may belong to the wrong speaker. We introduce **MESH**: **M**ulti-receiver **E**valuation of **S**patial **H**earing across Egocentric and Allocentric Views. The egocentric view is a receiver's

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[1]Evaluation code: https://github.com/Wenanzhi/MESH-Benchmark. Dataset: https://huggingface.co/datasets/Heiheihaha17/MESH-Benchmark.

self view: spatial queries and predictions use its local coordinate frame. The allocentric view instead uses a shared 3D world coordinate system. We define these views by their coordinate frames and specify their information inputs separately. In our evaluation, egocentric tasks use one receiver's recording with peer collaboration supplying robot poses, playback transcripts, and playback intervals; allocentric tasks use all available receiver recordings and metadata. These input protocols leave the system architecture open. An audio-language model may predict answers directly, whereas an agentic system may coordinate signal-processing tools and specialist models to obtain and combine perceptual evidence (Wang et al., 2026). Both must meet the same task requirements: a supplied transcript identifies a source to localize, a spatial cue identifies speech to transcribe, and a scene-level query requests source locations or trajectories.

We instantiate this setting with approximately 40,000 questions across eight static and five dynamic tasks. The dataset contains 800 base scenes spanning 100 rooms; each base scene produces one static and two independently sampled dynamic realizations that share speech content, noise, and initial layout. Synchronized first-order Ambisonic recordings capture spatial observations at each receiver, while prescribed source and receiver motion changes the geometry over time. Static and dynamic localization and spatial transcription are therefore evaluated within shared scenes using physical errors and target-specific word error rates. We evaluate eleven audio-model configurations and an agentic reference system. The best static distance-estimation accuracy is 21.90% within 20% relative error, and the best direction-tracking accuracy is 41.54% within 20°. The lowest word error rates across the four transcription tasks remain 59.15–73.10%. Analyses across source counts, acoustic conditions, and motion, supported by original model responses, reveal wrong-speaker transcription, inaccurate distance estimates, and nearly constant predictions despite changing source–receiver distances. These results identify limitations in the basic perception needed for scene understanding and motivate further study of how peer information and distributed observations can support reliable associations between speech and its physical source.

2 Related Work

Spatial audio evaluation. AIR-Bench, MMAU, and AudioBench evaluate broad audio-language capabilities (Yang et al., 2024; Sakshi et al., 2025; Wang et al., 2025). STAR-Bench assesses acoustic perception and spatio-temporal reasoning (Liu et al., 2026b), while RMS-AQA couples event grounding with spatial and temporal multiple-choice reasoning, including moving-source scenes (Chen et al., 2026). SpatialSoundQA and BiDepth use discretized spatial answers (Zheng et al., 2024; Biswas et al., 2026), whereas SpatialSceneQA, SO-Bench, and SEA-LM also include numerical spatial prediction (Liu et al., 2026a; Zhu et al., 2026; Kumar et al., 2026). ST-AudioQA evaluates trajectories in its encoder, while its language model primarily answers categorical, binned, and relational questions (Oh et al., 2026); we mark its trajectory coverage as partial. Table 1 compares numerical localization and trajectories, spatially cued transcription, joint multi-receiver input, and receiver motion. Full numerical coverage requires direct spatial estimates scored by physical error, rather than multiple-choice answers or coarse bins.

Distributed perception and target speech selection. LOCATA evaluates localization and tracking with source and array motion, reporting results for each array separately (Evers et al., 2020). L3DAS22 evaluates spatial speech enhancement and localization (Guizzo et al., 2022), while LibriWASN and CHiME-7 support speech processing across distributed devices (Schmalenstroeer et al., 2023; Cornell et al., 2023). DSS LibriSpeech evaluates localization-informed transcription (Tang et al., 2024), and DAE-TSE uses partial transcripts to cue target speaker extraction (Li et al., 2026). Here, joint multi-receiver input requires joint use of recordings from spatially separated devices observing the same scene, not merely multiple audio channels; receiver motion refers to movement of the recording device itself. MESH combines corresponding receiver views with shared-base static and dynamic scenes, using transcripts to identify speakers for numerical localization and spatial cues to specify whose complete input speech to transcribe.

3 MESH benchmark

MESH evaluates spatial speech understanding from two coordinate-based views: the **egocentric view** expresses spatial queries and predictions in a re-

Benchmark	Numerical spatial localization	Numerical trajectories	Spatially cued transcription	Joint multi-receiver input	Receiver motion
SpatialSoundQA (Zheng et al., 2024)	◐	✗	✗	✗	✗
BiDepth (Biswas et al., 2026)	◐	✗	✗	✗	✗
STAR-Bench (Liu et al., 2026b)	✗	✗	✗	✗	✗
SpatialSceneQA (Liu et al., 2026a)	✓	✗	✗	✗	✗
ST-AudioQA (Oh et al., 2026)	◐	◐	✗	✗	✗
SO-Bench (Zhu et al., 2026)	✓	✗	✓	✗	✗
SEA-LM Eval (Kumar et al., 2026)	✓	✗	✓	✗	✗
RMS-AQA (Chen et al., 2026)	◐	✗	✗	✗	✗
LOCATA (Evers et al., 2020)	✓	✓	✗	✗	✓
L3DAS22 (Guizzo et al., 2022)	✓	✗	✗	✓	✗
MESH	✓	✓	✓	✓	✓

Table 1: Comparison of spatial audio evaluation protocols. ✓: explicitly evaluated; ◐: discretized output or auxiliary evaluation; ✗: not evaluated in the public protocol. Comparison criteria are defined in Section 2.

ceiver's local frame, while the **allocentric view** uses a shared 3D world coordinate system. Both views address the same question: *who said what and where?* We define the views by their spatial reference frames and separately specify the recordings and collaborative information available in our evaluation.

We organize the benchmark around two task objectives: **localization**, which estimates where sound sources are, and **transcription**, which recovers what a spatially specified person said. Each objective is evaluated under applicable **static** and **dynamic** motion settings. Static tasks concern fixed source and receiver positions, while dynamic tasks require localization or transcription as spatial relationships change over time. Together, the two views and motion settings define 13 tasks. Figure 1 summarizes the benchmark: panel (a) combines speech, rooms, and noise into a shared scene; panel (b) shows the two coordinate views and their information inputs; panel (c) constructs recordings, labels, and questions; panel (d) characterizes the resulting acoustic conditions and spatial coverage. Table 7 in Appendix specifies the input cues and outputs for all tasks under both views.

3.1 Scene and Coordinate Views

Scene setup. Each scene contains $H \in \{1, 2, 3\}$ humans and $N \in \{1, 2, 3\}$ robot receivers. The figure uses H01, H02, etc. for humans and R01, R02, etc. for robot receivers. Robot i records a four-channel first-order Ambisonic (FOA) signal $X_i \in \mathbb{R}^{4\times L}$, where L is the number of samples per channel. FOA encodes omnidirectional pressure and three directional components at one receiver location. All receivers record over the same time interval. Speech sources include humans and robots playing speech. For each source s, c_s denotes its transcript, and $\mathbf{x}_s(t) \in \mathbb{R}^3$ denotes its world-frame position at time t. The known pose of robot i is $P_i(t) = (R_i(t), \mathbf{x}_i(t))$, where $\mathbf{x}_i(t)$ is the world-frame position of its local origin and $R_i(t) \in \mathbb{R}^{3\times3}$ is the rotation matrix describing its orientation. A point with robot-local coordinates $\mathbf{r}$ therefore has world coordinates $R_i(t)\mathbf{r} + \mathbf{x}_i(t)$.

Egocentric view and peer collaboration. The egocentric view is robot i's *self view*: source locations and spatial queries are expressed in its local coordinate frame, for example as a speaker's direction or distance relative to the robot. The relative position of source s is

$$\mathbf{r}_{is}(t) = R_i(t)^\top \left[\mathbf{x}_s(t) - \mathbf{x}_i(t)\right]. \tag{1}$$

In our evaluation, robot i observes its own recording X_i. Peer collaboration supplies robot metadata $M_j = (P_j, c_j^{\text{out}}, \tau_j)$, containing robot j's pose, speech playback transcript, and playback intervals. This information helps distinguish human speech from robot playback without sharing peer recordings or human speech annotations. Given public room context K and a query q_i, the prediction is

$$\hat{y}_i^{\text{ego}} = f_\theta^{\text{ego}}\left(X_i, \{M_j\}_{j=1}^N, K, q_i\right). \tag{2}$$

All spatial cues and predictions use robot i's local frame.

Allocentric view. The allocentric view represents source locations in a shared 3D world coordinate system as $\mathbf{x}_s(t)$, independent of any particular receiver's position or orientation. In our evaluation,

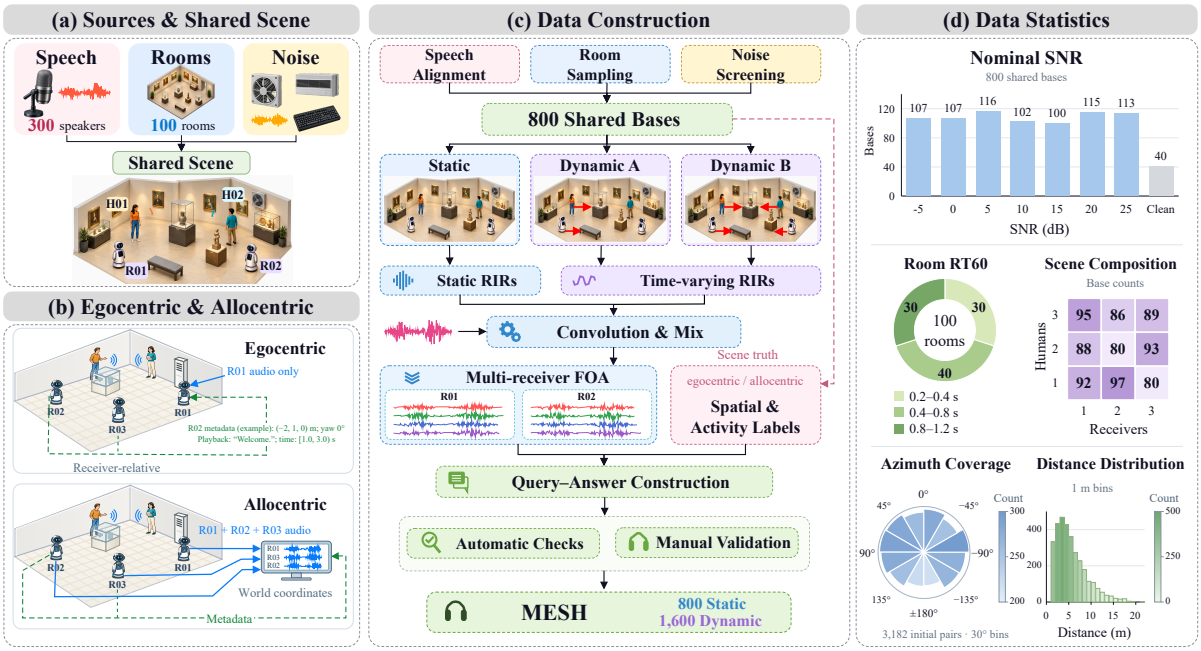

Figure 1: Overview of MESH: (a) scene sources, (b) coordinate views and available information, (c) construction and validation workflow, and (d) acoustic and spatial coverage. Panels are explained in Section 3; scene drawings and metadata examples are schematic.

it receives synchronized recordings and metadata from all available receivers. Given an allocentric query q^{allo}, the prediction is

$$\hat{y}^{\text{allo}} = f_\theta^{\text{allo}}\left(\{X_i, M_i\}_{i=1}^{N}, K, q^{\text{allo}}\right). \quad (3)$$

The system must associate acoustic observations across receivers with the same physical source. When $N = 1$, the two views use the same recording and robot metadata, but spatial cues and outputs remain in the receiver-local and world frames, respectively. The coordinate frame defines the view; the input protocol specifies the available information and leaves model architectures open.

Motion settings and queries. Static scenes keep source and receiver poses fixed. In dynamic scenes, humans, robots, or both may move. Source motion changes its world position, while receiver motion changes the observed direction and distance even for a stationary source.

Each task provides a query specifying the target or requested output. Localization queries may ask for all sources or use speech content or spatial cues to select a target. Transcription queries use spatial cues to select whose speech to recover. Corresponding queries can refer to the same physical source across receivers and views, with spatial cues expressed in the appropriate coordinate frame. Only annotations explicitly provided as query cues are visible; other human annotations are reserved for evaluation. Questions are evaluated independently, without scene images or access to other tasks' questions and answers.

Task identifiers. The codes in Table 3 specify view (E: egocentric; A: allocentric), motion setting (S: static; D: dynamic), and output (Dir: direction; Dist: distance; Pos: world position; Trans: transcription). An optional C marks a supplied transcript that selects the source for localization. Thus, ES-CDir is static egocentric transcript-conditioned direction localization. Dynamic localization returns a trajectory of the named quantity; tracking is not a separate objective. ED-Dist uses an initial spatial cue, not a transcript, so it has no C.

3.2 Egocentric Tasks

Egocentric tasks express spatial cues and predictions in the observing robot's local coordinate frame. They use the robot's own recording with peer metadata supplying additional spatial and speech information.

Static localization. *Direction localization* (ES-Dir) estimates the azimuth and elevation of all humans and other robots producing audible speech playback, given the source count. *Distance localization* (ES-Dist) selects one human through a unique azimuth sector and estimates that

person's distance. *Transcript-conditioned direction localization* (ES-CDir) and *Transcript-conditioned distance localization* (ES-CDist) instead use a supplied speech transcript to identify a human and estimate the speaker's direction or distance, respectively. The latter two tasks share the same transcript, target, and receiver, testing two spatial properties of a speaker identified by content.

Static transcription. *Spatially cued transcription* (ES-Trans) uses a person's direction and distance to select whose speech to transcribe. The system must recover all speech produced by that person during the input clip. Unlike localization, which predicts where a source is, transcription uses location to select the speech to recover.

Dynamic localization. Given H, the number of humans, direction trajectory localization (ED-Dir) estimates the azimuth and elevation of each person over time while maintaining consistent anonymous identities. Audio played by the robot is treated as interference. For distance trajectory localization (ED-Dist), a target person is identified using an initial speech interval and azimuth sector. The task is then to track that person's distance over time, even after they move outside the initial sector. Both tasks require consistent source identification as the spatial positions of the speakers change. *Distance trajectory localization* (ED-Dist) uses an initial speech interval and azimuth sector to identify one human, then estimates that person's distance over time, even after they leave the sector. Both tasks require maintaining the same source identity as spatial observations change.

Dynamic transcription. *Spatially cued transcription* (ED-Trans) provides two speech-active spatial anchors at least 1 s apart, each specifying direction, distance, or both. The anchors identify the same person, whose complete speech must be transcribed throughout the clip, including speech outside the anchor times.

3.3 Allocentric Tasks

Allocentric tasks express spatial cues and predictions in a shared 3D world coordinate system and use all available receiver recordings. They require associating acoustic observations across receivers and expressing spatial predictions in world coordinates.

Static localization. Given the human count H, *source-set position localization* (AS-Pos) estimates the world-frame positions of all humans as an unordered set of H points. *Transcript-conditioned position localization* (AS-CPos) instead uses a supplied speech transcript to identify one human and estimate that person's world position. Both tasks require three-dimensional localization, but differ in whether the target is selected by speech content.

Static transcription. *Region-cued transcription* (AS-Trans) selects a human through a supplied region in world coordinates and asks for that person's complete speech. If the region contains no human, the system must return an empty transcript.

Dynamic localization. *Transcript-conditioned trajectory localization* (AD-CPos) uses a supplied speech transcript to identify one human and requests that person's world-coordinate trajectory. The system must maintain the same source association as the person moves and acoustic observations change.

Dynamic transcription. *Trajectory-cued transcription* (AD-Trans) provides the target human's complete world trajectory, including silent periods but without speech-activity labels. The system must use the trajectory to identify the person across receiver recordings and transcribe all their speech throughout the clip.

3.4 Data Curation Pipeline

We combine recorded speech and environmental noise with acoustic simulation to obtain consistent source identities and spatial annotations across receivers and motion settings (Figure 1).

Shared scenes and acoustic rendering. Eight bases in each of 100 exhibition rooms define speech, robot playback, noise, initial geometry, and duration, up to 30 s. Speech is drawn from a LibriSpeech (Panayotov et al., 2015) inventory of 300 speakers; screened non-speech recordings and synthetic noise provide environmental interference. Each base produces one static and two independently sampled dynamic realizations (Dynamic A and B in Figure 1c), preserving content and initial layout. SoundSpaces 2.0 (Chen et al., 2022) renders FOA audio at every receiver through room impulse responses (RIRs), which describe sound propagation from each source to each receiver. Static RIRs or time-varying RIRs are convolved with the source signals and mixed with environmental noise. Each receiver's own playback is ideally removed; other robots' speech remains. Dynamic paths are

horizontal, with fixed height and robot yaw. The paired realizations support comparison of shared content, but are not single-factor motion interventions.

Queries and validation. Speech alignment supplies word and activity times; room sampling selects the scene geometry, and noise screening excludes unsuitable recordings. The scene truth branch in Figure 1c supplies source positions and speech-active intervals in each coordinate frame. These labels and transcripts generate view-specific queries and reference answers. Source identities link the same speaker across receivers; local coordinates are transformed for each egocentric view. Automated checks cover audio reconstruction, noise calibration, motion feasibility, coordinate consistency, and unique target selection after prompt rounding. Dynamic egocentric transcription candidates with insufficient relative motion are filtered before release. The manual validation stage supports human listening and annotation review. Appendix C.4 describes the implemented interface and procedure.

The resulting test set contains 800 static and 1,600 dynamic realizations, 4,761 receiver recordings, and approximately 40,000 questions across 13 tasks (Appendix C.2). Figure 1d reports nominal signal-to-noise ratio (SNR) over bases and room reverberation time (RT60, the time for sound energy to decay by 60 dB) over rooms. Environmental noise occurs in 95% of bases; nominal SNR spans -5 to 25 dB, while Clean means no added noise. Received ratios vary by target and receiver. Scene composition counts human–receiver population combinations; azimuth and distance plots count 3,182 initial human–receiver pairs once per base, showing spatial coverage rather than the question distribution. Appendix C.2 gives construction parameters, distributions, and validation scope.

4 Experiments

4.1 Setup and Evaluation Protocol

Systems and inputs. We evaluate eleven audio-model configurations and MESH-agent on 19,117 static and 20,882 dynamic questions. The audio models include Qwen3-Omni (Xu et al., 2025) (FOA-W and FOA-sep), Spatial-Omni-7B and Spatial-Omni-30B (Zhu et al., 2026), JAEGER Neural and Classical (Liu et al., 2026a), BAT (Zheng et al., 2024), MiDasheng-Spatial (Hu et al., 2026), Qwen3.5-Omni-Plus (FOA-W and FOA-sep), and Qwen3.8-Omni-Flash. MESH-agent uses a frozen Qwen3-14B (Yang et al., 2025a) planner to coordinate specialist audio models and signal-processing tools. FOA-W uses the omnidirectional channel, while FOA-sep supplies WXYZ as separate audio clips. Qwen3.8-Omni-Flash receives native four-channel FOA input. All systems use only the recordings, metadata, and queries permitted by each task.

Evaluation metrics. Spatial predictions are considered correct within 20° direction error, 20% relative distance error, or 2 m world-position error. Dynamic localization predicts absolute spatial values on a 0.5-s grid and is evaluated on speech-active frames, with consistent source assignment across direction tracks. Missing or invalid predictions count as failures. Spatial transcription is evaluated by corpus word error rate (WER), which reflects both target selection and speech recognition. The 800 empty-region queries are evaluated separately by empty-answer accuracy.

Overall score. We also report a question-level Overall score for each view, giving equal weight to localization and transcription. Let $\mathcal{L}_v$ and $\mathcal{S}_v$ denote their respective question sets in view v, including both static and dynamic tasks:

$$\mathrm{Overall}_v = 50\left(\frac{\sum_{i\in\mathcal{L}_v} a_i}{|\mathcal{L}_v|} + \frac{\sum_{i\in\mathcal{S}_v} s_i}{|\mathcal{S}_v|}\right). \quad (4)$$

Here a_i is the fraction of correct targets or active frames within question i, and $s_i = \max(0, 1 - \mathrm{WER}_i)$ for nonempty transcripts, with WER expressed as a ratio. For empty-region queries, s_i is one for a correct empty answer and zero otherwise. Questions are equally weighted within each group, and the resulting score ranges from 0 to 100. We retain the original task metrics for detailed comparisons.

Analysis protocol. We analyze performance by human count, received SNR and SIR, source distance, and motion. The condition analysis covers six model configurations: Qwen3-Omni (FOA-sep), Spatial-Omni-7B, Spatial-Omni-30B, MiDasheng-Spatial, Qwen3.8-Omni-Flash, and Qwen3.5-Omni-Plus (FOA-sep). We further compare three models under different peer-metadata and receiver-recording inputs on a fixed 100-room subset. These comparisons preserve the same ques-

		Static					Dynamic			
System	Input	ES-Dir	ES-Dist	ES-CDir	ES-CDist	ES-Trans	ED-Dir	ED-Dist	ED-Trans	Overall
Metric		Acc↑	RelAcc↑	Acc↑	RelAcc↑	WER↓	Acc↑	RelAcc↑	WER↓	
Single-channel encoding LALMs										
Qwen3-Omni	FOA-W	11.34	18.79	10.15	20.52	96.53	3.34	20.73	128.00	20.56
Qwen3-Omni	FOA-sep	13.07	19.61	8.71	**21.68**	93.55	4.07	19.52	123.70	21.41
Qwen3.5-Omni-Plus	FOA-W	18.13	21.34	9.37	20.77	80.74	6.56	**22.24**	78.00	24.75
Qwen3.5-Omni-Plus	FOA-sep	17.33	**21.90**	9.15	20.65	72.14	7.64	19.40	**73.10**	26.52
Multichannel encoding LALMs										
BAT	Binaural	0.00	0.00	0.00	0.00	99.89	0.00	0.00	99.82	0.07
MiDasheng-Spatial	Binaural	4.62	6.10	7.95	13.76	77.21	0.00	1.20	86.56	13.23
Spatial-Omni-7B	FOA	7.63	6.91	10.97	15.02	87.68	1.30	0.70	97.35	13.66
Spatial-Omni-30B	FOA	18.91	5.81	24.76	5.88	137.11	6.70	5.75	165.46	18.54
JAEGER Neural	FOA	13.62	7.23	7.54	3.33	101.62	0.00	3.35	101.20	2.80
JAEGER Classical	FOA	7.60	0.06	0.19	0.00	103.47	0.04	3.14	100.75	0.94
Qwen3.8-Omni-Flash	FOA	41.37	13.98	**50.03**	19.55	72.09	**41.54**	19.36	73.32	30.18
Agentic specialist tool-calling system										
MESH-agent	FOA	**49.34**	16.47	47.17	20.43	**61.09**	28.61	19.06	78.98	**34.97**

Table 2: Egocentric results (%). Task codes follow Table 3. Bold: column best. Overall uses Eq. 4.

		Static			Dynamic		
System	Input	AS-Pos	AS-CPos	AS-Trans	AD-CPos	AD-Trans	Overall
Metric		Acc↑	Acc↑	WER↓	Acc↑	WER↓	
Single-channel encoding LALMs							
Qwen3-Omni	FOA-W	1.19	0.31	104.00	6.48	95.10	13.77
Qwen3-Omni	FOA-sep	1.19	0.06	109.59	6.45	92.06	14.22
Qwen3.5-Omni-Plus	FOA-W	9.49	7.62	**71.06**	6.93	74.10	19.45
Qwen3.5-Omni-Plus	FOA-sep	10.99	11.37	72.65	6.41	68.91	21.86
Multichannel encoding LALMs							
BAT	Binaural	0.00	0.00	99.89	0.00	99.98	0.04
MiDasheng-Spatial	Binaural	7.00	7.56	83.36	0.19	81.18	15.36
Spatial-Omni-7B	FOA	11.62	6.43	96.39	0.28	89.87	12.55
Spatial-Omni-30B	FOA	2.87	3.31	150.43	2.66	152.76	12.72
JAEGER Neural	FOA	5.00	3.56	106.95	0.00	101.57	6.58
JAEGER Classical	FOA	0.00	0.06	102.87	0.00	101.38	1.21
Qwen3.8-Omni-Flash	FOA	3.75	4.43	73.98	4.19	73.68	15.90
Agentic specialist tool-calling system							
MESH-agent	FOA	**32.04**	**31.79**	78.54	**18.39**	**59.15**	**31.01**

Code	Full task name
Egocentric: Static	
ES-Dir	Direction localization
ES-Dist	Distance localization
ES-CDir	Transcript-conditioned direction localization
ES-CDist	Transcript-conditioned distance localization
ES-Trans	Spatially cued transcription
Egocentric: Dynamic	
ED-Dir	Direction trajectory localization
ED-Dist	Distance trajectory localization
ED-Trans	Spatially cued transcription
Allocentric: Static	
AS-Pos	Source-set position localization
AS-CPos	Transcript-conditioned position localization
AS-Trans	Region-cued transcription
Allocentric: Dynamic	
AD-CPos	Transcript-conditioned trajectory localization
AD-Trans	Trajectory-cued transcription

Table 3: Allocentric results (%). Full task names on the right are grouped by coordinate view and motion setting. Bold: column best. Overall uses Eq. 4.

tions and targets, and use room-clustered bootstrap intervals to assess performance differences.

Further details on model configurations, MESH-agent, scoring, and experimental analyses are provided in Appendix A and B.

4.2 Overall Performance

Tables 2 and 3 report results on eight egocentric and five allocentric tasks, respectively, covering both static and dynamic settings. Spatial tasks use threshold accuracy (Acc) or relative-distance accuracy (RelAcc), while transcription tasks use word error rate (WER).

Localization. Qwen3.8-Omni-Flash leads the direct audio models on egocentric direction tasks, reaching 50.03% on ES-CDir and 41.54% on ED-Dir. Yet its distance accuracies remain below 20%, and its allocentric localization accuracies below 5%. MESH-agent performs better on allocentric localization, reaching 32.04% on AS-Pos and 18.39% on AD-CPos. These results show that accurate direction estimates do not necessarily translate into accurate distances or world positions.

Transcription. The best WERs across the four transcription tasks remain between 59.15% and 73.98%. Qwen3.5-Omni-Plus leads several direct-model transcription tasks, while MESH-agent achieves the lowest WER on ES-Trans (61.09%) and AD-Trans (59.15%). MESH-agent reaches Overall scores of 34.97 and 31.01 in the egocentric and allocentric views, respectively. No direct model consistently leads across tasks, and better speech recognition does not guarantee better spatial grounding. Additional model comparisons are provided in Appendix B.2.

4.3 Ablations and Diagnostics

This section highlights the key insights from our ablation studies and diagnostic analyses. Detailed evaluation results and additional examples are provided in Appendices B.2, B.3 and D.

Peer information provides limited benefits. Removing peer metadata reduces Qwen3-Omni (FOA-sep)'s ES-Dir accuracy from 14.20% to 9.34%, but improves some Spatial-Omni-7B results. The effects vary across models and tasks, with no consistent improvement from additional peer information. Current systems do not reliably use shared robot metadata to improve spatial understanding.

Distributed recordings offer selective gains. Using all receivers improves dynamic allocentric transcription for some models. For example, Qwen3-Omni (FOA-sep)'s WER decreases from 87.44% to 82.54%. Spatial-Omni-7B shows a larger improvement, but most of its gain comes from a few length-limited responses. Spatial localization shows no consistent benefit from additional recordings. Thus, access to distributed acoustic observations does not yet lead to reliable improvements across tasks.

Competing sources challenge spatial transcription. Figure 2 shows that transcription errors increase with the number of humans and stronger speech interference. More importantly, models often transcribe the wrong speaker. For Qwen3.5-Omni-Plus (FOA-sep), 25.68% of positive transcription answers match another human's speech more closely than the target's. Models also produce speech for empty target regions. These failures show that recognizing speech and identifying its physical source remain distinct challenges.

Distance and motion remain poorly grounded. Models frequently underestimate distant sources.

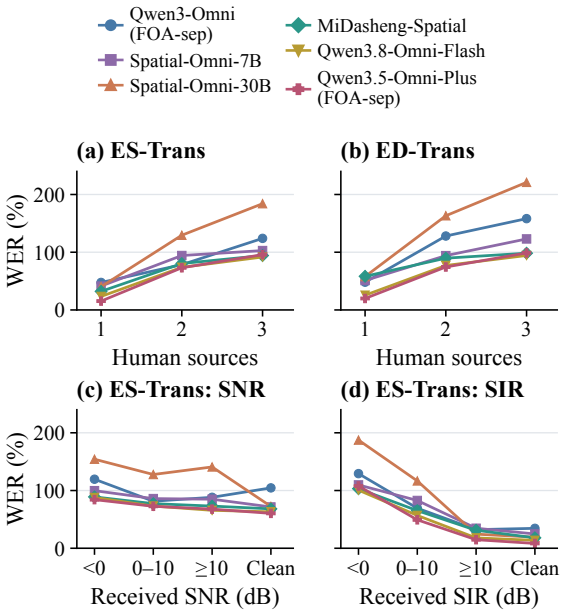

Figure 2: Egocentric transcription WER. Clean denotes no added noise (c) or no competing speech (d).

For targets at least 5 m away, Spatial-Omni-7B and Spatial-Omni-30B predict a median distance of only 1.5 m, compared with approximately 7.6 m in the ground truth. Their predicted distance trajectories also remain nearly constant in over 97% of eligible cases, even when the true distance changes. As illustrated in Figure 7(b), producing a trajectory does not necessarily mean tracking the source's motion.

5 Conclusion

We introduce MESH to evaluate spatial speech understanding through egocentric and allocentric views, covering 13 localization and spatial transcription tasks under static and dynamic conditions. Our results reveal persistent difficulties in associating speech with the correct physical source, particularly in crowded scenes and under motion. More importantly, peer information and distributed recordings provide only selective improvements, with no consistent gains in spatial localization. These findings highlight a gap between the availability of spatial information and the ability of current systems to use it effectively. Future models should better integrate peer metadata, receiver geometry, and acoustic observations across locations, while maintaining source identity over time. We release the dataset, construction pipeline, and evaluation code to support the development of collaborative and distributed spatial hearing systems.

Limitations

Data coverage. MESH uses simulated exhibition-room mixtures and English read speech. Generalization to spontaneous conversations, other languages, and physical recording environments remains untested. Motion is restricted to horizontal paths with fixed height and robot yaw, and long silent intervals are underrepresented. RIR interpolation approximates time-varying propagation without exact Doppler modeling, while recorded noise may retain its original acoustics. Overlap with model pretraining data cannot be ruled out. The preliminary listening audit covers only a small subset of scenes and does not establish cross-receiver consistency.

Observation assumptions. Recordings are synchronized, receiver poses and playback logs are known, and each receiver's own playback is ideally removed. Source counts and target-selection cues are supplied where required. Clock drift, uncertain calibration, missing recordings, and active receiver control are not evaluated. Offline evaluation does not establish real-time performance or embedded deployment feasibility.

Experimental scope. Condition analyses cover selected models and do not independently control acoustic factors. Information ablations cover three models on a subset of scenes; removing metadata also changes prompt length. The reported confidence intervals do not account for multiple comparisons. MESH-agent was evaluated on a release with prior results, rather than an untouched test set, and the contribution of agentic planning has not been isolated on the full benchmark. The observed errors do not uniquely identify their internal causes.

Ethical Considerations

Potential risks. Spatial speech understanding can support assistive interfaces and robots in shared environments, but may also enable monitoring or speaker tracking without appropriate consent. Incorrect speech–source associations can attribute an utterance to the wrong person. Benchmark performance does not establish suitability for surveillance, personal identification, or consequential decisions about individuals. Deployment requires appropriate consent, privacy safeguards, and validation in the intended environment.

Data and scope. The benchmark uses existing speech and noise recordings in simulated acoustic scenes. Scene-local target identifiers do not anonymize the underlying voices, and data reuse remains subject to upstream licenses. English read speech and simulated rooms do not establish generalization across languages, demographic groups, or physical environments. Acoustic rendering and model inference require computational resources; subsequent analyses reuse existing predictions where possible to avoid unnecessary repeated inference.

Human review. A preliminary listening audit involved seven participants and produced 57 completed review records covering 56 physical realizations. The audit included 36 pilot and 21 formal reviews. Its limited coverage does not establish inter-rater agreement or cross-receiver consistency. The review procedure and observations are documented in Appendix C.4.

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A Detailed Experimental Configurations

Table 4 summarizes the evaluated interfaces. This appendix retains settings needed to interpret the comparisons; the release archive contains full manifests, raw outputs, prompts, and detailed tables.

A.1 LALM Inputs and Decoding

Qwen3.8-Omni-Flash and Qwen3.5-Omni-Plus are closed-source API models. The other audio models use open-source releases. FOA-W supplies the omnidirectional channel; FOA-sep encodes WXYZ as four labeled mono inputs. This does not establish native joint FOA processing. Spatial-Omni-7B and Spatial-Omni-30B retain their spatial and semantic frontends. BAT and MiDasheng-Spatial receive KEMAR binaural decodings, without independent channel normalization. KEMAR measurements are described by Gardner and Martin (1995). Receiver labels and original clip timing are retained.

The frozen Qwen3-Omni, Spatial-Omni-7B, JAEGER Neural, and BAT workers use greedy decoding, temperature zero, seed zero, and BF16 language inference. The output budget is 1,024 tokens for static and transcription outputs and 8,192 for trajectories, subject to model context limits. Spatial-Omni-7B uses labeled temporal blocks of at most 20 s; BAT uses 32-kHz blocks of at most 10 s. These blocks are not additional microphones. MiDasheng-Spatial uses `mispeech/midashenglm-spatial` revision `160ae51b51be`, official code `e12bc10b1279`, FP32 vLLM 0.22 (Kwon et al., 2023), greedy decoding, and repetition penalty 1.05 with the same token budgets. Its full clips retain the 185-sample filter tail without extending the task grid. KEMAR decoding differs from its ARI training distribution, and multi-receiver blocks extend its native evaluation setting; these differences limit attribution.

JAEGER checkpoints. JAEGER Neural and JAEGER Classical use the authors' released `neural_iv/audio_only_2` and `classical_iv/audio_only_2` checkpoints, respectively, from the JAEGER repository. OWL is excluded because its release does not establish a reproducible match to the final CoT-trained model; a historical ST-AudioLM run is excluded because its FOA basis conversion was incorrect.

A.2 Input and Output Audits

Native reference checks match six Spatial-Omni-30B inputs, eight MiDasheng-Spatial inputs, and six JAEGER Classical feature tensors. Short decoder checks validate those examples, not all full-length answers.

Additional archived inference settings. The archived Qwen3-Omni worker configurations use tensor parallelism of one, a maximum context length of 32,768 tokens, and a batched-token budget of 8,192. Prefix caching is enabled, with at most 128 sequences and a GPU memory utilization setting of 0.96. The multimodal input limits allow up to three audio blocks for FOA-W and twelve

System	Input	Recorded configuration
Qwen3-Omni	FOA-W, FOA-sep	Qwen3-Omni-30B-A3B-Instruct; BF16, greedy; 1,024 or 8,192 output tokens
Qwen3.5-Omni-Plus	FOA-W, FOA-sep	Closed-source API; W only or four separate WXYZ inputs
Spatial-Omni-7B	FOA	Spatial-Omni-7B and SO encoder; labeled blocks up to 20 s; BF16, greedy
Spatial-Omni-30B	FOA	Native spatial and semantic frontends; input and short-decoder reference checks
JAEGER Neural	FOA	Neural IV, audio-only checkpoint 2; BF16, greedy
JAEGER Classical	FOA	Classical IV, audio-only checkpoint 2; native input-feature audit
BAT	Binaural	KEMAR conversion; 32-kHz, labeled blocks up to 10 s; BF16, greedy
MiDasheng-Spatial	Binaural	KEMAR conversion; full clips; FP32, greedy, repetition penalty 1.05
Qwen3.8-Omni-Flash	FOA	Closed-source API; native four-channel FOA input
MESH-agent	FOA	Qwen3-14B planner; eight steps, four logical acoustic calls, 14,000 input tokens

Table 4: Recorded input and inference configurations.

for FOA-sep, each containing at most 480,000 samples. These are interface limits: the loader supplies only the recordings permitted by the current query. The Spatial-Omni-7B records identify the Spatial-Omni-7B and SO-Encoder weight files by SHA-256, separately from the Qwen2.5-Omni-7B text decoder.

A.3 Agentic Specialist Tool-Calling System

MESH-agent denotes frozen Agent V2, with Qwen3-14B (Yang et al., 2025b) revision `40c069824f42` as its planner. Capability cards and request-local memory guide structured calls; the planner selects observations and tool transcripts rather than inventing numerical observations or rewriting transcripts. The tools include S5, OV-Seld (Shimada et al., 2025), the Jiang distance estimator, DiCoW (Polok et al., 2026), Spatial-Omni distance alternatives, Qwen3-ASR-1.7B (Shi et al., 2026), direction-conditioned FOA MVDR/cardioid filtering, and coordinate transforms. Source slots and activity are predicted, not supplied by reference enrollment. Each question permits eight planner steps, four logical acoustic calls, and 14,000 input tokens per call. A logical call may invoke several models or receivers and is not a neural forward-pass count. The policy was frozen on development data; cached observations and recorded execution passed deterministic replay checks. Earlier evaluation on this release makes this a frozen regression, not an untouched-test evaluation. Cached replay also does not measure complete online latency.

Capability cards and execution contract. The frozen system uses three task-specific card sets: the static `v13` cards, the temporal `exhibition-temporal-v1-public-protocol` cards, and the `exhibition-dev-v2-manual` cards for ES-CDist, ES-Trans, ED-Trans, and AD-Trans. Each card specifies its purpose, applicable tasks, prerequisites, argument schema, returned evidence, and limitations. The V2 extension exposes source observation, source-conditioned recognition, spatially conditioned transcription, mixture observation, source-range refinement, range estimation, candidate inspection, answer selection, and failure reporting. Available tools and candidate references are restricted by the task and the current observation state. Arguments are checked against JSON schemas before execution; invalid calls produce feedback rather than silently repaired observations. Candidate transcripts remain evidence rather than instructions, and memory is confined to the current question. Terminal selection returns an observed candidate value without rewriting its transcript.

Planner runtime. Qwen3-14B runs in BF16 with vLLM 0.10.2 and tensor parallelism of two, without quantization or CPU offloading. Generation uses temperature zero, seed 290930, and at most 192 output tokens per planner call, with thinking disabled. This output budget applies to the structured tool call; a selected transcript is subsequently returned by the executor. The server context limit is 16,384 tokens, while each planner input is limited to 14,000 tokens. Calls use the native Hermes tool format with JSON-schema-constrained decoding. The decoder omits the unsupported `uniqueItems` grammar constraint, while the full prompt schema and executor retain the uniqueness requirement.

Acoustic execution and caching. The additional Qwen3-ASR-1.7B backend uses BF16 inference with greedy decoding, English as the requested language, an empty text context, and a maximum output length of 512 tokens. It processes complete predicted source waveforms or direction-conditioned recordings; reference transcripts are not supplied

as recognition prompts. The full evaluation used 40 RTX 4090 GPUs, running the additional ASR in 40 shards before launching 20 two-GPU planner replicas. Existing specialist observations were reused from frozen caches. Source- and cue-conditioned ASR outputs were sealed before fresh planner inference, and scoring followed prediction freezing and deterministic replay checks. This execution arrangement evaluates answer quality using cached perception and does not establish live audio-to-answer latency.

B Detailed Results and Diagnostic Analyses

This appendix gives the metric definitions and diagnostic evidence behind the main-text findings on source competition, spatial errors, and selective use of additional information.

B.1 Metrics and Reporting Scope

For unit directions $\hat{u}, u$, distances $\hat{r}, r > 0$, and positions $\hat{x}, x$, the respective errors are

$$e_\theta = 180 \arccos(\mathrm{clip}(\hat{u}^\top u, -1, 1))/\pi, \quad (5)$$

$$e_r = |\hat{r} - r|/r, \qquad e_p = \|\hat{x} - x\|_2. \quad (6)$$

Thresholds follow Section 4.1. Source sets use one-to-one matching; ED-Dir uses one assignment throughout the clip. Direction and world-position scores pool reference positions, using speech-active frames for tracks. ED-Dist first averages active-frame success within each question, then weights questions equally. Missing predictions fail. Coverage is the fraction of references with an evaluable prediction, using the same aggregation. Missing directions receive 180° diagnostic error; distance and position errors condition on valid predictions. Thus, low conditional error alone need not imply success.

Corpus WER is $100 \sum_i (S_i + D_i + I_i) / \sum_i N_i$, using the person selected by the query. Failed positive answers incur complete deletion; repetitions and insertions remain. The 800 empty-region queries have no WER denominator and instead require a valid empty answer; refusal is not absence. Equation 4 averages localization accuracy within each question and clips transcription scores *before* averaging, $s_i = \max(0, 1 - (S_i + D_i + I_i)/N_i)$, or the correct-empty indicator. This differs from clipping corpus WER or averaging task means. The weights were adopted after results were available and fixed across models. Computations retain the frozen record-level denominators behind the published scale of 40,000 questions.

View	Localization	Semantic	Overall
Egocentric	27.85	42.10	34.97
Allocentric	23.67	38.35	31.01

Table 5: MESH-agent Overall components (0–100; higher is better), including empty targets.

Empty targets and Agent components. Correct absence affects rankings: excluding empty regions would change Allocentric Overall from 14.22 to 16.10 for Qwen3-Omni (FOA-sep) and from 15.36 to 14.19 for MiDasheng-Spatial. MESH-agent's independently checked question-level archive uses the same protocol; its localization and semantic components are shown in Table 5. Its empty-answer accuracy is 22.50%.

Qwen3.5-Omni-Plus components and uncertainty. The main tables report both interfaces' Overall scores. Correct-empty accuracy is 6.00% (FOA-W) and 12.13% (FOA-sep). The FOA-sep-minus-FOA-W Overall differences have room-bootstrap intervals $[1.40, 2.15]$ and $[1.95, 2.91]$ points, respectively. Against Flash, FOA-sep is lower Egocentrically (-3.66; $[-4.35, -3.01]$) and higher Allocentrically ($+5.96$; $[5.33, 6.59]$). These intervals describe scene sampling for the evaluated systems.

B.2 Condition Strata and Failure Distributions

Condition definitions. The human count H is the number present, not simultaneous speakers; scenes with $H = 1$ can still contain robot playback. SNR compares target W-channel power with environmental noise; SIR compares it with summed competing speech, including waveform cross-terms. Powers use emitted target-active samples, after ideal self-playback cancellation, before conversion to dB. Finite bins are < 0, $[0, 10)$, and ≥ 10 dB. In Figure 2, Clean means no environmental noise in (c), and no competing speech in (d); the connecting lines show ordered conditions, not a continuous numerical interval to Clean. The no-noise group has 134 static questions from 40 bases in 34 rooms; the SIR ≥ 10 dB group has 72 questions from 43 rooms. Different group compositions preclude a universal monotonicity or causal interpretation.

Selected differences use 2,000 room-cluster bootstrap replicates (Field and Welsh, 2007), recomputing corpus WER from summed edits and reference words, with paired room multiplicities and seed 20261012. The 95% percentile intervals are unadjusted for multiple comparisons and reflect scene sampling, not reruns. All twelve three-human-minus-one-human intervals in the main figure are positive. The SIR contrasts are also consistently adverse, while SNR and noise-type rankings differ by model. Full contrasts and denominators are retained in the archive.

Transcription outcomes. Figure 3 uses 12,925 positive-reference questions per configuration. Priority labels are mutually exclusive: no usable transcript; non-transcription answer; repetition; other-source match; mixed-source evidence; exact or near target match; other overlong output; unresolved content error. They are diagnostic categories, not a causal decomposition. Classical and MESH-agent lack comparable full profiles. An other-source match requires at least five words, candidate WER at most 35%, target WER at least 65%, and a gap of at least 30 points. Human reference transcripts are hidden, but robot playback texts are public, so robot-text matches do not identify listening versus copying. Repetition requires at least 100 words, ten occurrences of one trigram, and a distinct-trigram ratio below 0.35. Spatial-Omni-30B's 275 flagged answers (2.13%) contribute 49.21% of its edits. This differs from reaching a token limit.

Distance and motion. Far-target medians condition on a valid scalar and true distance at least 5 m. Nearly constant ED-Dist outputs have predicted span below 0.01 m, at least three readable active distances, and reference span above 0.5 m. This excludes missing or short outputs: Spatial-Omni-7B and Spatial-Omni-30B have 1,576 and 4,176 eligible questions, of which 98.54% and 97.49% are constant. Flash has 670 constant outputs among 4,537 eligible questions (14.77%); Plus FOA-sep has 1,125 among 4,035 (27.88%). Their ED-Dist accuracies remain 19.36% and 19.40%, respectively. Matched static–dynamic comparisons average dynamic variants within each base–human–receiver group, then weight groups equally. Flash's 2,999 matched groups show a 4.76-point increase in mean question WER (95% room interval 3.57–5.93); changing cues prevent a motion-only claim.

B.3 Controlled Information Comparisons

The paired study selects one base per room by fixed hash before scoring: 5,090 questions across 100 rooms, with 28,217 requests per model across variants. All 84,651 requests are complete. Qwen3-Omni (FOA-sep), Spatial-Omni-7B, and Spatial-Omni-30B retain fixed adapters, greedy decoding, and token budgets. Each egocentric variant has 3,714 questions. E0 keeps full inputs; E1 removes peer poses; E2 playback text; E3 playback times; E4 all playback; E5 peer poses and playback. Deletions affect both structured inputs and prompts, retaining recording, calibration, query, and target. Shorter prompts are a confound; no length-matched placebo is included. Allocentric variants retain all public metadata while changing only recordings. The control is fixed R01, selected before scoring; fusion contrasts require at least two receivers. Singleton, pair, and full sets retain receiver bindings. Primary contrasts are E0 versus E5 and all receivers versus R01, with 2,000 paired room-bootstrap replicates and unadjusted 95% intervals. These subset scores do not enter full-test Overall.

The main text reports both benefits and harms from metadata deletion. For the 270 paired AD-Trans questions, Qwen3-Omni improves on 65, worsens on 38, and ties on 167 with all receivers; neither input reaches the token limit. Spatial-Omni-7B improves on 93, worsens on 71, and ties on 106, but six length-limited answers account for 71.44 of its 77.68-point WER reduction. These decompositions retain every answer and the full denominator. The separate earlier Spatial-Omni-30B study uses 12,891 questions per context; all 16 contrast intervals contain zero. Its different population and newly measured baseline are not mixed into the paired subset above.

Geometric opportunity versus acoustic evidence. On 135 AS-CPos questions, intersecting true source bearings perturbed by 5-degree RMS noise improves Acc@2m from 47.56% for a fixed-R01 ray with a 5-m depth prior to 89.07% with all rays (difference interval 32.82–50.25 points). This is an oracle-bearing diagnostic, not an acoustic baseline. A mean FOA-intensity frontend gated by true target-exclusive activity improves only from 8.89% to 11.85% on the same full population (interval -2.96–9.52), counting missing rays as failures. Only 70 questions have sufficient exclusive activity at every receiver. The contrast identifies geometric potential without isolating acoustic estima-

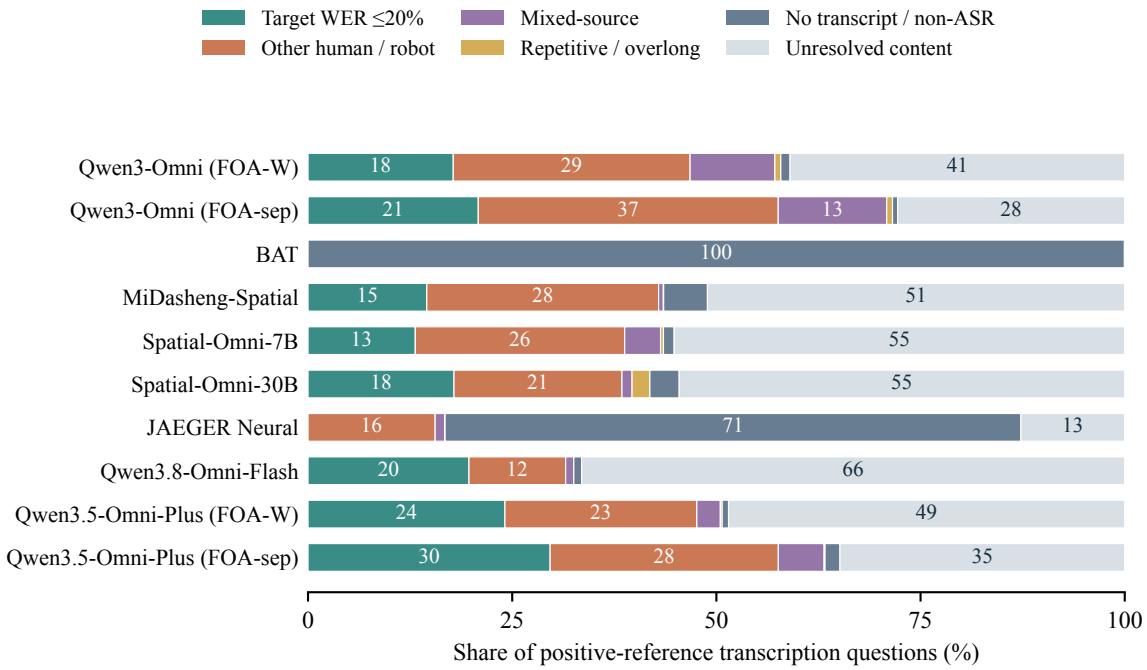


Figure 3: Transcription outcome distribution over 12,925 positive questions per configuration. Mutually exclusive priority labels are grouped for display; the text defines their limits.

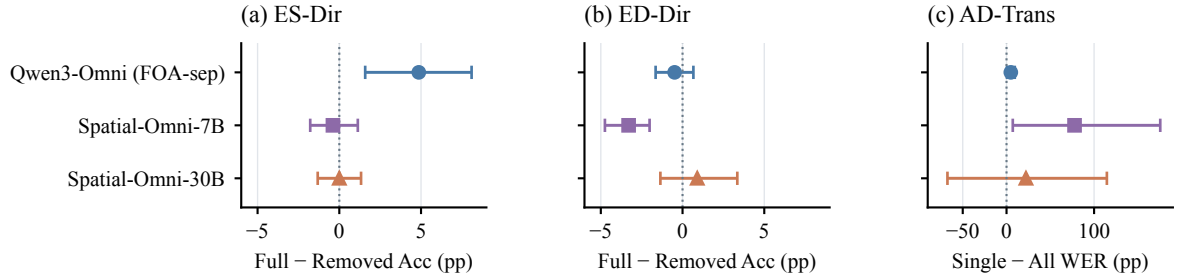


Figure 4: Selected paired information effects with 95% room-bootstrap intervals. Positive values favor keeping metadata (a,b) or adding recordings (c).

tion, source association, and fusion as independent causes of model failure.

B.4 Separate Agent Development Comparison

A later round uses 1,520 tuning questions in 12 rooms and 992 holdout questions in eight rooms. Three candidate changes to memory, verification, and capability cards fail the selection criteria, so V2 remains frozen. The fixed MVDR cascade outperforms it on all four measured holdout tasks (Table 6). The holdout retains 44 interrupted requests as failures (948 complete returns), so it does not establish uninterrupted generalization. This development population is separate from the main evaluation. The four displayed tasks use 1,120 tuning and 720 holdout questions; the remaining 400 and 272 questions provide regression checks.

C Implementation, Prompts, and Human Review

C.1 Task Contracts and Coordinates

Table 7 gives all task inputs and outputs using the main-text codes. Dataset identifiers remain unchanged: `edge` and `cloud` mean egocentric and allocentric. A FOA bundle is one microphone location; temporal chunks are not additional receivers. Directions use azimuth zero in front, positive left, and elevation positive up. World $+y$ is up; zero robot yaw faces $-z$ and positive yaw turns toward

Task	Metric (%)	Tuning DEV		Development holdout	
		MESH-agent	Fixed MVDR	MESH-agent	Fixed MVDR
Static egocentric ES-CDist	RelAcc@20%↑	21.92	23.85	19.19	20.35
Static egocentric ES-Trans	WER↓	98.29	87.35	116.16	91.69
Dynamic egocentric ED-Trans	WER↓	106.81	86.29	94.77	92.03
Dynamic allocentric AD-Trans	WER↓	81.79	70.65	82.64	69.90

Table 6: Separate development comparison using the same specialist observations. The fixed MVDR cascade makes no planner calls.

View	**Motion Setting**	**Task**	**Input**	**Required Output**
Egocentric	Static	ES-Dir	Human and peer-playback count	Direction set
		ES-Dist	Unique human azimuth sector	Target distance
		ES-CDir	Target transcript	Target direction
		ES-CDist	Target transcript	Target distance
		ES-Trans	Target direction and distance	Transcript
	Dynamic	ED-Dir	Human count H	H direction tracks
		ED-Dist	Initial speech interval and azimuth sector	Target distance track
		ED-Trans	Two local spatial anchors	Transcript
Allocentric	Static	AS-Pos	Human count H	Set of H world positions
		AS-CPos	Target transcript	Target position
		AS-Trans	World-coordinate region	Transcript
	Dynamic	AD-CPos	Target transcript	Target position track
		AD-Trans	Target world trajectory on a 0.5-s grid	Transcript

Table 7: Tasks by view and motion setting. Input lists cues added to the recordings and public context in Eqs. 2–3. Transcript means all target speech in the clip.

$+x$. Receiver poses transform world displacements into local front–left–up coordinates. Angles are degrees; distance and position are meters. Spatial tracks contain every requested 0.5-s timestamp and a speech-activity flag; active frames carry spatial values, inactive frames omit them. ED-Dir returns H tracks with stable anonymous IDs, and AS-Pos returns an unordered set of H world positions. AD-Trans receives a full world trajectory including silence, without activity labels, and returns the full transcript.

C.2 Construction and Validation

The benchmark scale is 40,000 questions drawn from a 300-speaker LibriSpeech inventory, with 800 bases in 100 rooms, 800 static and 1,600 independently sampled dynamic realizations, and 4,761 FOA recordings. Within a base, realizations share speech crops, playback, noise, initial layout, and duration (mean 11.71 s); different bases use distinct recordings. R01 is silent; other receivers may play speech, with ideal self-playback removal. Signals are 16-kHz float32 ACN/SN3D FOA $[W, Y, Z, X]$. Of 800 bases, 760 add noise (722 recorded and 38 synthetic); 40 are clean. Noise gain is calibrated at static H01–R01 and retained across realizations, so received target SNR and SIR vary. Common peak scaling preserves relative gains. Prescribed horizontal minimum-jerk paths span 1.5–5 m at candidate peak speeds 0.4–0.8 m/s, with fixed height and robot yaw. RIR knots every 0.2 s and two-time interpolation retain propagation delays and reverberation tails. Automated checks cover audio, labels, timestamps, prompt bindings, target correspondence, and unique selection after coordinate rounding. Filtering removes 1,424 ambiguous dynamic transcription candidates while retaining their recordings. Oracle and missing-output tests check scoring. These checks do not establish listening agreement or real-device generalization.

ES-Trans: actual query

Select the same HUMAN speaker satisfying ALL of the following spatial conditions. Throughout this static recording: direction [azimuth,elevation] [-164,11] degrees, spherical angular tolerance 10 degrees; distance 2.2 m, tolerance +/-0.5 m. Exactly one human satisfies this query. Ignore other people, robot playback and environmental noise. Transcribe ALL of that person's speech in the ENTIRE input. Return only the English transcript; no timestamps, speaker IDs, JSON fields or explanation.

Figure 5: A frozen task query, after shared public context.

C.3 Prompts and Answer Extraction

The loader supplies the current query, permitted recordings and robot metadata; other questions, scene images, and evaluator-only human labels are hidden. The task table specifies the variable cues inserted into each prompt. The following example preserves the actual wording of the frozen ES-Trans query; full prompts for all thirteen tasks and receiver bindings are archived. Gold-blind extraction removes answer wrappers without correcting predicted words or numbers. WER uses the requested target, never the best-matching scene speaker. The frozen English tokenizer lowercases and uses its original word-pattern rule; numbers are not expanded. Repetition is retained. Spatial repair recovers existing fields only. One inherited manual extraction remains hash-checked in the archive and is not an exhaustive listening review. MESH-agent submits its selected tool transcript without rewriting.

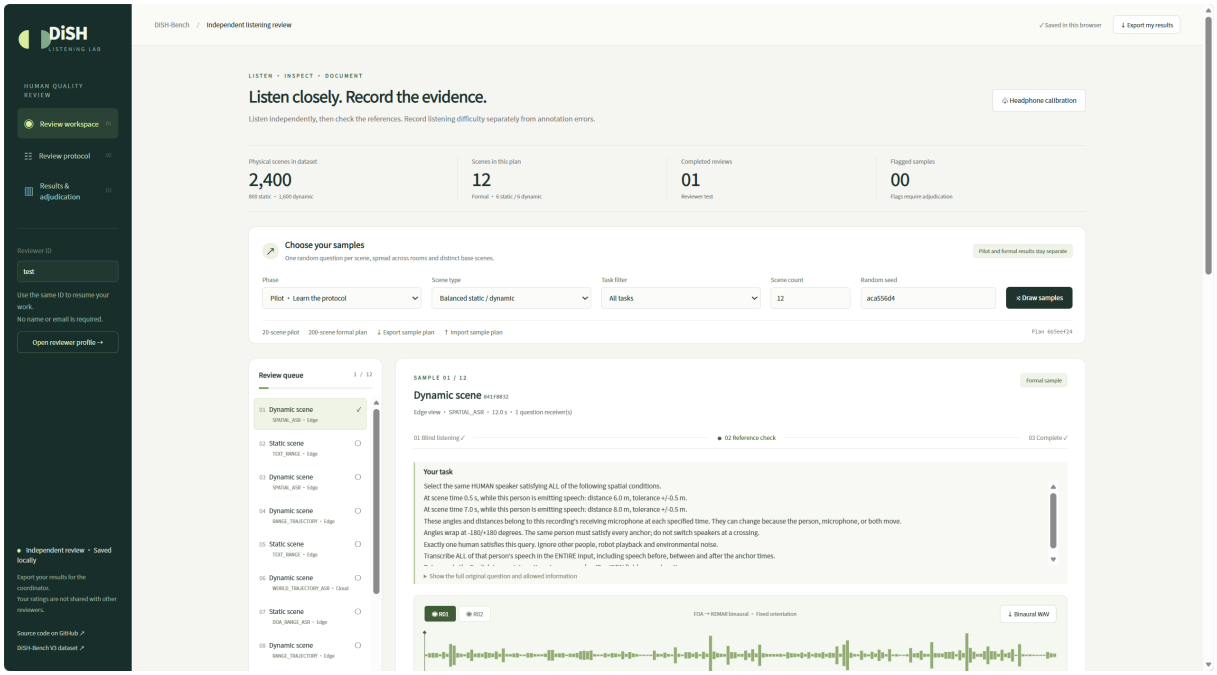
C.4 Human Review Interface and Procedure

A reproducible review record should distinguish the presented recordings and query, the unedited model response, the extracted answer, and the review decision. Figure 6 shows the listening and reference-check views of this interface.

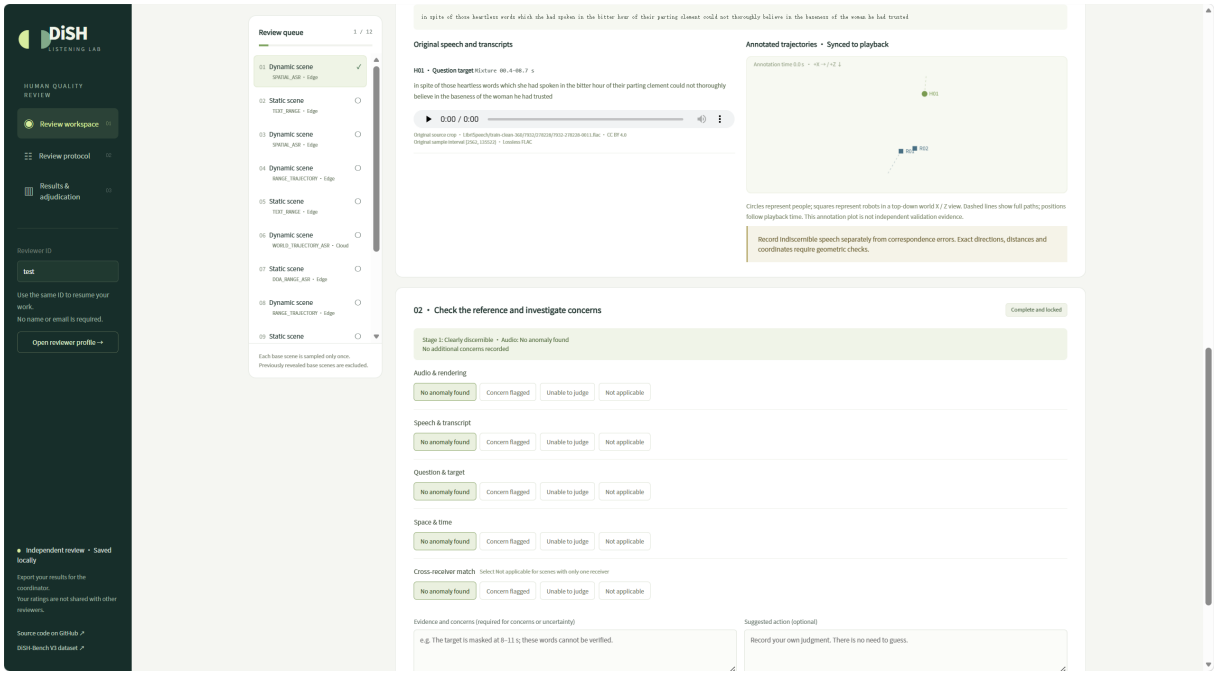
Review interface. We implement a browser-based interface for two-stage dataset quality review. It presents the original question and its permitted receiver recordings, with waveform navigation, receiver switching at a shared playback time, and timestamped annotations. Four-channel ACN/SN3D FOA is decoded to binaural audio using a KEMAR first-order FIR filter bank. Playback preserves channel relationships and uses a common gain across receivers, with headphone and left/right calibration controls. The first stage records target audibility and suspected rendering anomalies before reference answers are revealed. The interface requires at least 95% playback coverage for each task-bound receiver before a completed first-stage judgment can be locked; playback failures are recorded separately. After locking, the second stage reveals the reference answer, original source speech and transcripts, and trajectories synchronized with playback. Reviewers separately assess rendering quality, speech–text correspondence, question–target correspondence, spatial–temporal continuity, and cross-receiver correspondence.

Sampling and review records. The interface supports reproducible sampling at the physical-realization level, selecting one question per realization with fixed random seeds, static/dynamic quotas, and room-wise sampling without repeated base scenes. Exported plans fix the scene and question identities so that reviewers can independently inspect the same material. Pilot and formal records are kept separate. Ratings distinguish suspected annotation errors from masking and other legitimate acoustic difficulty, and retain uncertainty, non-applicability, comments, and affected time intervals. Results are saved locally and explicitly exported for aggregation. For shared samples, the analysis interface computes raw agreement and nominal Krippendorff's α from independent ratings; adjudication is stored separately without replacing those ratings. Pilot records, prior reference exposure, and playback failures are excluded from formal agreement analysis.

Listening audit. We collected 57 completed scene–question reviews covering 56 physical realizations, comprising 36 pilot and 21 formal reviews. We report the two phases separately. Before reference disclosure, 33 pilot and 18 formal samples were rated clearly discernible; each subset contained two partly discernible samples and one indiscernible sample. Three pilot reviews flagged initial audio-quality concerns, whereas none were flagged in the formal subset at this stage. After reference disclosure, formal reviews flagged no concerns about speech–text correspondence, question–target correspondence, or spatial–temporal continuity. Two formal records contained concerns about excessive background noise and perceived overlap-related distortion, respectively. One pilot record received concerns across all five review fields. These concerns remain unadjudicated.



(a) Sampling and listening workspace.



(b) Reference inspection and review form.

Figure 6: The listening-review interface. (a) Reviewers configure the sampling plan, navigate the review queue, inspect the question, and select a receiver recording. (b) The reference-check view places source speech and transcripts beside a playback-synchronized trajectory display. Separate controls record concerns about audio rendering, speech–transcript correspondence, question–target correspondence, space and time, and cross-receiver matching, with fields for supporting evidence and suggested actions. The trajectory display visualizes dataset annotations rather than independent geometric validation.

Scope. Each scene–question pair received a single review, precluding inter-rater agreement estimates. Formal playback logs covered only one receiver per question, so the submitted cross-receiver ratings do not establish cross-receiver consistency. The findings provide preliminary evidence about perceptual accessibility and potential quality issues in the reviewed subset.

D Qualitative Case Studies

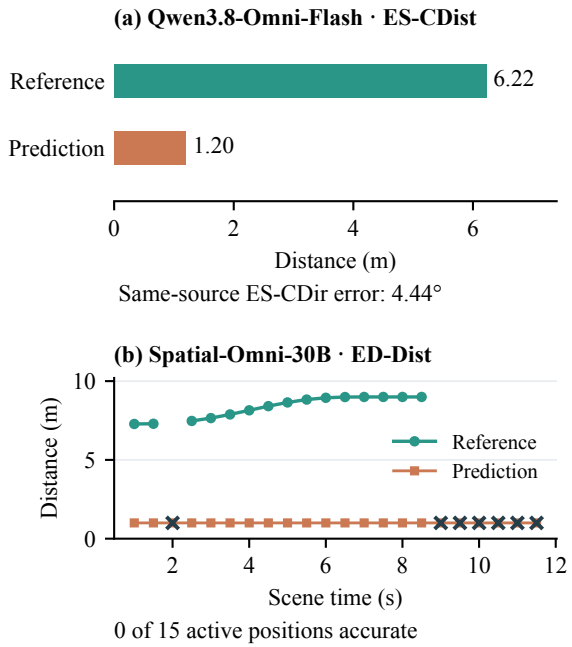


Figure 7: Direction can be accurate while distance is wrong (a); a well-formed track can ignore motion (b). Crosses indicate false speech activity.

Figure 8 compares all eleven direct configurations on the same two queries. For ES-Trans, several models reproduce the wrong human's sentence; for AS-Trans, several transcribe speech outside an empty requested region. JAEGER Neural returns a correct empty answer in the latter case. These examples distinguish intelligible recognition from spatial target selection. Responses are frozen original outputs, excerpts are marked, and scores use complete extracted answers; examples do not estimate prevalence.

MESH-agent on the shared queries. For the same ES-Trans example in Figure 8, MESH-agent incurs 21 word edits against a 17-word reference, corresponding to 123.53% WER. For the AS-Trans example with an empty target region, it returns a nonempty answer, yielding 27 insertion errors and failing the correct-empty criterion. Both answers pass the output-format check. These per-question records establish answer-level failures but do not distinguish errors in acoustic evidence acquisition, transcription, and final target selection; that attribution requires the corresponding tool observations and decisions.

(a) ES-Trans · Which person spoke?

Request · R01 audio
Direction [−65°, 11°], distance 2.6 m. "Transcribe ALL of that person's speech in the ENTIRE input."

Reference · evaluator only
he was now renting a former photographic studio on the top of a house in government street

Qwen3-Omni (FOA-W) · WER 105.9%
So that the children could look into her happy eyes and see the glow of her […]

Qwen3-Omni (FOA-sep) · WER 105.9%
So that the children could look into her happy eyes and see the glow of her […]

BAT · WER 100.0%
Yes

MiDasheng-Spatial · WER 105.9%
so that the children could look into her happy eyes and see the glow of her […]

Spatial-Omni-7B · WER 100.0%
so that the children could look into her happy eyes and see the glow of short […]

Spatial-Omni-30B · WER 105.9%
So that the children could look into her happy eyes and see the glow of her […]

JAEGER Neural · WER 100.0%
I'm sorry, but I can't provide the transcription of the speech as it's not included in […]

JAEGER Classical · WER 100.0%
I'm sorry, but I don't have the actual audio file to transcribe the speech. Could you […]

Qwen3.8-Omni-Flash · WER 105.9%
So that the children could look into her happy eyes and see the glow of her […]

Qwen3.5-Omni-Plus (FOA-W) · WER 105.9%
so that the children could look into her happy eyes and see the glow of her […]

Qwen3.5-Omni-Plus (FOA-sep) · WER 105.9%
So that the children could look into her happy eyes and see the glow of her […]

(b) AS-Trans · Is the target absent?

Request · R01 + R02 audio
WORLD box [4.2, 1.3, −1.2] to [4.7, 1.7, −0.7] m. Return the selected transcript, or an empty string if no match.

Reference · evaluator only
"" (empty string); the sole human is outside the queried region.

Qwen3-Omni (FOA-W) · incorrect
"Almost immediately after Major Vernon's departure, Henry Dunbar rang the bell for the servant who acted […]

Qwen3-Omni (FOA-sep) · incorrect
"Almost immediately after Major Vernon's departure, Henry Dunbar rang the bell for the servant who acted […]

BAT · incorrect
Yes

MiDasheng-Spatial · incorrect
Almost immediately after Major Vernon's departure, Henry Duffour rang the bell for the servant who acted […]

Spatial-Omni-7B · incorrect
"Almost immediately after Major Byerley's departure, Henry Duffer rang the bell for the servant, who acted […]

Spatial-Omni-30B · incorrect
Almost immediately after Major Vernon's departure, Henry Duffar rang the bell for the servant who acted […]

JAEGER Neural · correct empty
""

JAEGER Classical · incorrect
"Ah, I'm not sure what you're asking me to do here. I don't have any speech […]

Qwen3.8-Omni-Flash · incorrect
Almost immediately after Major Vernon's departure, Henry Dunbar rang the bell for the servant who acted […]

Qwen3.5-Omni-Plus (FOA-W) · incorrect
Almost immediately after Major Vernon's departure, Henry Dunbar rang the bell for the servant who acted […]

Qwen3.5-Omni-Plus (FOA-sep) · incorrect
Almost immediately after Major Vernon's departure, Henry Dunbar rang the bell for the servant who acted […]

Figure 8: The same two spatial queries across eleven direct configurations. Request boxes summarize the supplied cues; response boxes quote raw answers, with [. . .] marking truncation. Scores use complete extracted answers. Samples: `00453216fc112c99fd735b2c` (a) and `00afedbe599a7a8dfd77cd0d` (b).